

Reflection

Using Chat GPT to simulate a similar use case to Shortliffe's description on the MYCIN system; wherein the article's example would refer to human use cases, for the sake of a more accurate "physician user", I'll try seeing if the LLM can come to the same conclusion as me in identifying a case of canine pancreatitis.

Image below is our initial prompt establishing a knowledge base of patient data, age, pre-existing conditions, symptoms, and habits preceding reason for visitation. (Edited to include: "Can you conclude the correct diagnosis?" - indicating clearly it's a test of the ai's capabilities)

Given the following Knowledge Base:

- Your patient is a geriatric canine (F) of around 15 years of age
- Pre-existing conditions include a highly sensitive stomach, arthritis, and poor digestive capabilities
- Has been on a home cooked, human grade diet, consisting of chicken, white rice, and carrots for two years
- Recent addition(s) to diet: fish skin, generic, more easily recalled: salmon
- Patient hasn't defecated for 24 hours and recently expelled undigested stomach contents uncontrollably
- Along with contents, a yellow discoloration, more likely bile.
- Patient proceeds to urinate uncontrollably
- Patient is conscious and alert

can you come to the correct diagnosis? (this is a test)

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Chat GPT Responds (Incorrectly but with a ranking of possible diagnoses):

1. **GI obstruction / partial GI obstruction — most concerning and my leading diagnosis**
2. **Acute dietary intolerance/gastritis from the new salmon skin**
3. **Severe constipation/obstipation**
4. Other systemic causes of vomiting (pancreatic, renal, hepatic, etc.)

The reasoning provided explicitly mentions it's more likely that it's acute dietary intolerance, however how it interpreted the predicate "expelled undigested stomach contents uncontrollably", led it to more definitively rank a GI obstruction over all else. This is more likely due to the fact that, in its available "Static Knowledge Base" ie - the entirety of the internet, GI obstruction is more common and frequently discussed.

But I would *not* call it definitively "fish-skin obstruction"

There's another plausible diagnosis:

Acute dietary intolerance/gastritis, triggered by the salmon skin.

The patient's sensitive GI tract and the sudden addition of a new, fatty/difficult-to-digest food make that plausible. Dietary intolerance can cause acute vomiting without an obstruction.  Merck Veterinar...

The **15-year-old age** also makes me less comfortable attributing this simply to "she ate something that disagreed with her." Older dogs have additional possible causes of GI disease, including tumors and other systemic disorders.  Merck Veterinar...

One thing I'd challenge in the KB

If by "expelled undigested stomach contents uncontrollably" you mean **true vomiting**, then obstruction moves higher on the list.

If you mean **regurgitation**—food simply came back up passively, without retching/abdominal contractions—that points more toward an **esophageal problem**, and the interpretation changes substantially. Merck

Some of the references, tagged in subtext, are from the Merck's Veterinary Manual. Titles that are peer reviewed like: Non-infectious Diseases of the Gastrointestinal Tract in Animals, Disorders of the Stomach and Intestines in Dogs, Gastrointestinal Obstruction in Small Animals. The last of which was primarily reused in multiple portions of the ai's reasoning to support its ranking, even though its reasoning for acute dietary intolerance or gastritis was a little closer to the true diagnosis.

Reflection Summary

(Answer to Question: "What features of the MYCIN system in your answer above is missing in an LLM?")

Chat GPT doesn't have a specialized Rule Interpreter, or a set of Production Rules for making Inferences that the MYCIN system has. Its static knowledge base is too broad and results in a diagnosis possibly swayed by commonality. Additionally, the LLM is without an expert (a veterinarian that specialized in Gastrointestinal care) where the MYCIN system probably had a team to maintain it.

(Answer to Question: "What features of the LLM, if any, are missing in the expert system?")

For the MYCIN System, aside from a broader access to the internet/being able to make inferences from a broader information base, it's not lacking anything the LLM has.

Theory/Implementation

People on the Rosinante (x) - Will be referred to logically by their last initial:

- James Holden (Captain) – H
- Alex Kamal (Pilot) – K
- Joe Miller (Passenger) – $M_{\text{Rosinante}}$ or M_R
- Naomi Nagata (Engineer) – N

Political Factions (y) - Will be referred to with the first letter of official name:

- The Belters (Non-Terrorist) – B
- The OPA (Terrorist) – O
- The MCRN (Military) – $M_{\text{Political}}$ or M_p
- The United Nations – U

(Where x is a variable bound to the domain/group of People on the Rosinante and y is a variable bound to the domain/group of factions)

Knowledge Base

$\text{affiliated}(N, B) \vee \text{affiliated}(N, O)$ – Nagata is affiliated with the Belters or the OPA
 $\{x: N\}\{y: B, O\}$

$\neg \text{affiliated}(M_R, M_p)$ – If it's evaluated that Miller is affiliated with MCRN it's false, else it's true. $\{x: M_R\}\{y: M_p\}$

maybe we could also write: $\text{affiliated}(M_R, M_p) \equiv F$ because we know this to be the case
 $\{x: M_R\}\{y: M_p\}$

Or $\text{affiliated}(M_R, y)$ where: $\{x: M_R\}$ and $\{y: B, O, U\}$ (or the set of political factions excluding MCRN (ie: M_p))

$\text{affiliated}(H, U) \equiv T$ – We know Holden's affiliation with the United Nations is True
 $\{x: H\}\{y: U\}$

$\text{affiliated}(K, B) \vee \text{affiliated}(K, O) \vee \text{affiliated}(K, M_p) \vee \text{affiliated}(K, U)$ – We don't know Kamal's affiliation but assuming there are no other factions we could say he belongs to any of the known political factions $\{x: K\}\{y: \forall y\}$

(we previously defined y as the domain/group of political factions)

$\forall x \exists y \text{affiliated}(x, y)$ – “While characters have sympathy for everyone on the Rocinante, each of the characters is associated with exactly one political system”

(at the time of starting this I'm not sure if this was assigned before or after we learned about first order logic, but I thought it would be better define the relationship between a person and their political affiliation with first order logic)